

9. Write a Java program for a simple calculator using GridLayout.

Type	Input	Expected Output	Explanation
Success	10 + 5	15	Addition
Success	20 - 8	12	Subtraction
Success	6 * 7	42	Multiplication
Failure/Edge	10 / 0	Division by zero error	Invalid operation
Failure/Edge	Operator pressed without number	Error message	UI input validation

Algorithm:

1. **Start** the program.
2. Create a JFrame with two input fields, four operation buttons (+, -, *, /), and a result field.
3. Register ActionListener for all four buttons.
4. When a button is clicked, read the two numbers from the input fields.
5. Check whether both input fields contain values.
6. Perform the selected operation:
 - o + → Addition
 - o - → Subtraction
 - o * → Multiplication
 - o / → Division
7. Check for **division by zero** before performing division.
8. Display the result in the result field.
9. If the input is invalid, display an error message.
10. **Stop** the program when the window is closed.

Program:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class SimpleCalculator extends JFrame implements ActionListener {

    JTextField num1Field, num2Field, resultField;
    JButton addBtn, subBtn, mulBtn, divBtn;

    public SimpleCalculator() {
        setTitle("Simple Calculator");
        setSize(400, 250);
```

```

setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

setLayout(new GridLayout(4, 2, 10, 10));

add(new JLabel("First Number:"));
num1Field = new JTextField();
add(num1Field);

add(new JLabel("Second Number:"));
num2Field = new JTextField();
add(num2Field);

addBtn = new JButton("+");
subBtn = new JButton("-");
mulBtn = new JButton("*");
divBtn = new JButton("/");

addBtn.addActionListener(this);
subBtn.addActionListener(this);
mulBtn.addActionListener(this);
divBtn.addActionListener(this);

JPanel buttonPanel = new JPanel(new GridLayout(1, 4));
buttonPanel.add(addBtn);
buttonPanel.add(subBtn);
buttonPanel.add(mulBtn);
buttonPanel.add(divBtn);

add(buttonPanel);

add(new JLabel("Result:"));
resultField = new JTextField();
resultField.setEditable(false);
add(resultField);

setVisible(true);
}

```

```

@Override
public void actionPerformed(ActionEvent e) {

    try {
        if (num1Field.getText().trim().isEmpty() ||
            num2Field.getText().trim().isEmpty()) {

            resultField.setText("Error: Enter both numbers");

```

```

        return;
    }

    double num1 = Double.parseDouble(num1Field.getText());
    double num2 = Double.parseDouble(num2Field.getText());

    double result = 0;

    if (e.getSource() == addBtn) {
        result = num1 + num2;
    } else if (e.getSource() == subBtn) {
        result = num1 - num2;
    } else if (e.getSource() == mulBtn) {
        result = num1 * num2;
    } else if (e.getSource() == divBtn) {

        if (num2 == 0) {
            resultField.setText("Error: Division by zero");
            return;
        }

        result = num1 / num2;
    }

    resultField.setText(String.valueOf(result));

    } catch (NumberFormatException ex) {
        resultField.setText("Error: Invalid input");
    }
}

public static void main(String[] args) {
    new SimpleCalculator();
}
}

```